

Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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Summary

Dynamic Data Scientist & Machine Learning Engineer with a proven track record in launching groundbreaking AI and ML data products, amassing over \$1.3 billion in transformative business value. Expertise spans comprehensive end-to-end analytics, adept stakeholder engagement, inspiring mentorship, and cutting-edge MLOps.

Professional Experience

Lead Data Scientist - Caterpillar Inc. - Surprise, AZ

09/2022 to Present

- Architected and delivered end-to-end data products in Python, Snowflake, and R (including an AI platform that predicts equipment repair times) driving an \$80M reduction in warranty costs and earning a CEO Award
- Productionized models on Amazon SageMaker using deep learning (PyTorch, TensorFlow), gradient boosting (XGBoost, LightGBM), and LLM/sentence-transformer embeddings; improved repair-time prediction accuracy by 40% and overall model performance
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication

Reliability Analytics Manager - Amazon.com - Tempe, AZ

02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis, and wrote a white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Leveraged Amazon SageMaker for deep learning using PyTorch and TensorFlow, employing XGBoost and LightGBM for gradient boosting to enhance model accuracy. Also used AWS Redshift and PostgreSQL to improve data management, accessibility, and processing speed
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager - Empire-Cat - Mesa, AZ

01/2021 to 02/2022

- Developed a time-series AI model and MLOps pipeline using SQL, R, and Python for demand forecasting, which identified key supply chain risks
- Led a team in designing, building, and maintaining scalable data infrastructure, enhancing Empire's data-driven capabilities
- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days

Reliability Engineering Manager - Empire-Cat - Mesa, AZ

01/2016 to 01/2021

- Supervised 8 engineers and technical communicators and acted as a customer-facing expert on mining equipment. Mentored up to 6 interns per year
- Deployed reproducible research with R and SQL to track product issues and incorporate reliability analysis, which became the template for Caterpillar to incorporate in their enterprise solution
- Led the root cause failure analysis program for Empire using a data, facts, and evidence approach, which supported the FMEA and 6 Sigma process at Caterpillar

- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding \$1.2 billion in savings over 6 years
- Led a team of 5 Technical Communicators in a client-facing role to effectively convey customer feedback to Caterpillar's engineering organizations, driving improvements in product development and customer satisfaction
- Developed condition monitoring application with R and SQL which allowed fluids sample analysis on thousands of products, which saved Empire \$100,000 in software licensing
- Partnered on design of experiments to improve reliability, safety and durability
- Developed Python and MS-SQL application to track performance and outcomes of prototype designs tested in the field

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate • Deep Learning with PyTorch, Keras and Tensorflow Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **NCEES:** Passed FE Exam
- **US Department of Labor:** Operating Engineer Vocational Certificate

Education

Bachelor of Science in Mechanical Engineering - Arizona State University - Tempe, AZ 06/2003 to 05/2006
STEM, Multivariate Calculus, Differential Equations, Linear Algebra, Numerical Methods, Algorithms, Data Structures, Statistics, Mechanics, Material Science, Manufacturing, Finite Element Analysis, Computational Fluid Dynamics, Internal Combustion Engines, Mechanisms, Heat Transfer, Thermodynamics, Manufacturing, Economics, Cost, Finance, Control Systems, Design

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn